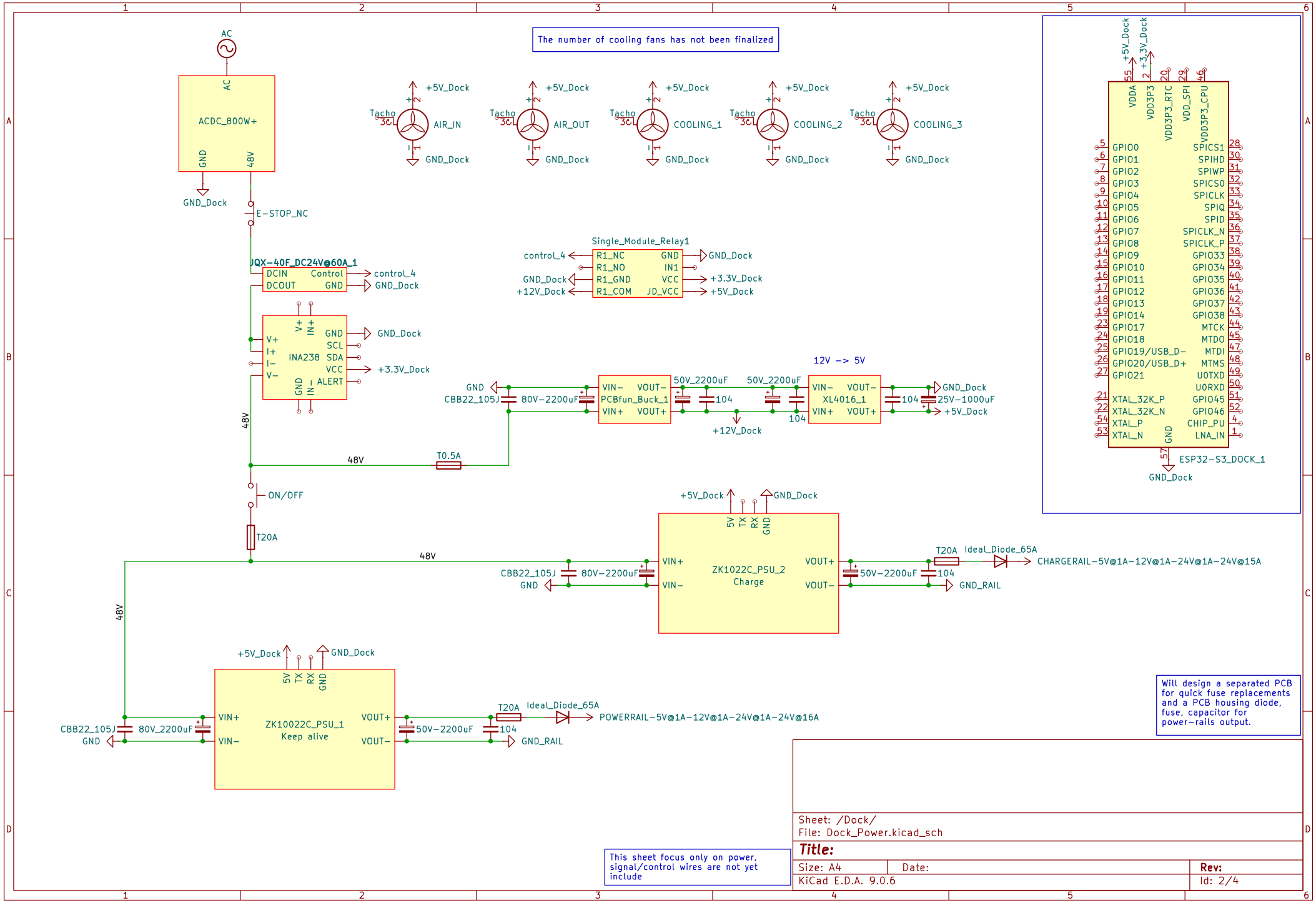
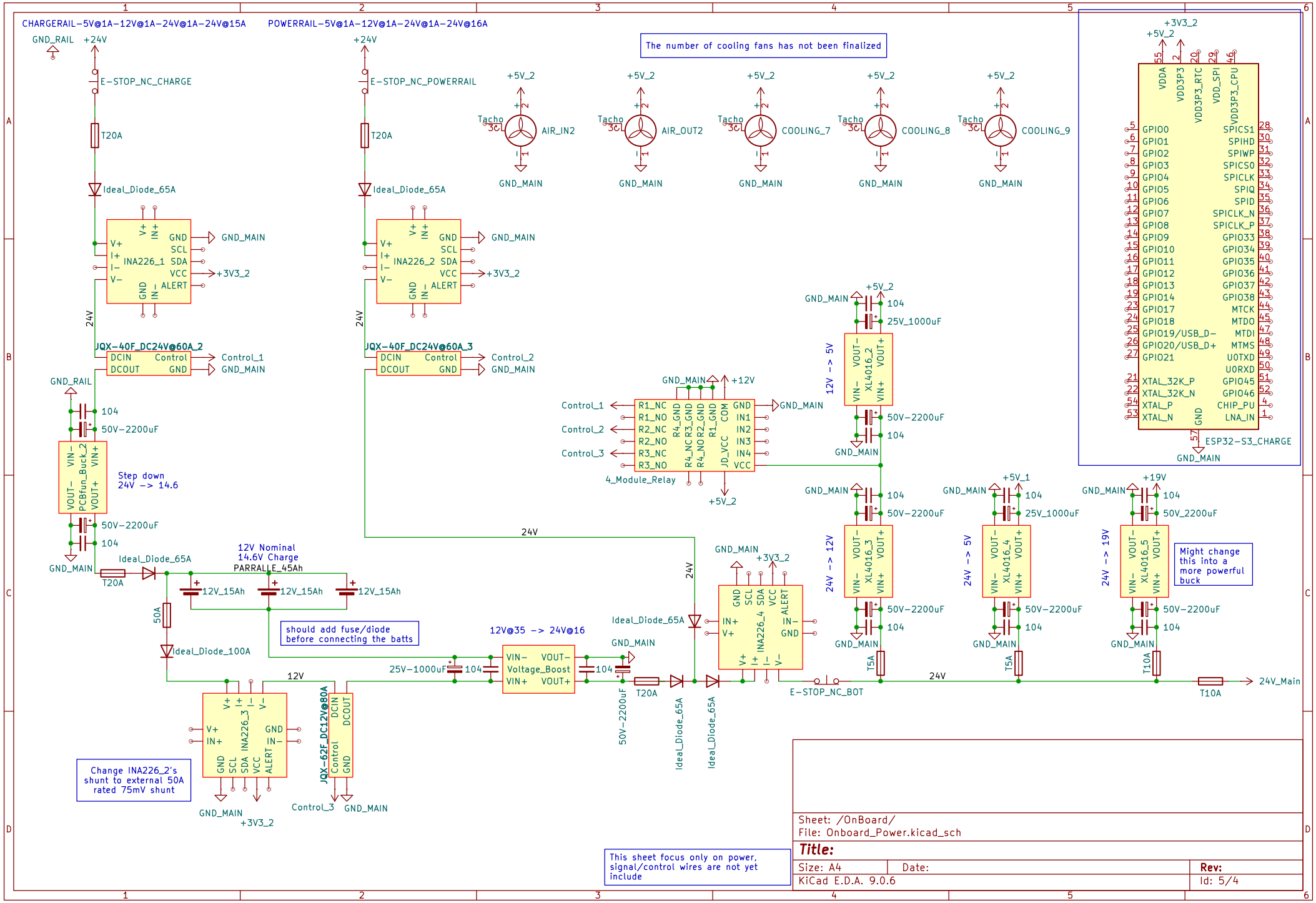

The number of cooling fans has not been finalized
AC
ACDC_800W+
GND
48V
GND_Dock
E-STOP_NC
JQX-40F_DC24V@60A_1
DCIN
Control
DCOUT
GND
control_4
GND_Dock
INA238
V+
I+
I-
V-
SCL
SDA
VCC
ALERT
+3.3V_Dock
Single_Module_Relay1
R1_NC
R1_NO
R1_GND
R1_COM
GND
IN1
VCC
JD_VCC
+12V_Dock
+5V_Dock
AIR_IN
AIR_OUT
COOLING_1
COOLING_2
COOLING_3
Tacho
12V -> 5V
CBB22_105J
80V-2200uF
PCBfun_Buck_1
VIN-
VIN+
VOUT-
VOUT+
50V_2200uF
104
XL4016_1
25V-1000uF
T0.5A
ON/OFF
T20A
ZK1022C_PSU_2
Charge
5V
TX
RX
50V-2200uF
Ideal_Diode_65A
CHARGERAIL-5V@1A-12V@1A-24V@1A-24V@15A
GND_RAIL
ZK10022C_PSU_1
Keep alive
POWERRAIL-5V@1A-12V@1A-24V@1A-24V@16A
ESP32-S3_DOCK_1
VDDA
VDD3P3
VDD3P3_RTC
VDD_SPI
VDD3P3_CPU
+3.3V_Dock
GPIO0
GPIO1
GPIO2
GPIO3
GPIO4
GPIO5
GPIO6
GPIO7
GPIO8
GPIO9
GPIO10
GPIO11
GPIO12
GPIO13
GPIO14
GPIO17
GPIO18
GPIO19/USB_D-
GPIO20/USB_D+
GPIO21
XTAL_32K_P
XTAL_32K_N
XTAL_P
XTAL_N
SPICS1
SPIHD
SPIWP
SPICS0
SPICLK
SPIQ
SPID
SPICLK_N
SPICLK_P
GPIO33
GPIO34
GPIO35
GPIO36
GPIO37
GPIO38
MTCK
MTDO
MTDI
MTMS
U0TXD
U0RXD
GPIO45
GPIO46
CHIP_PU
LNA_IN
Will design a separated PCB for quick fuse replacements and a PCB housing diode, fuse, capacitor for power-rails output.
This sheet focus only on power, signal/control wires are not yet include
Sheet: /Dock/
File: Dock_Power.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 9.0.6
Id: 2/4


CHARGERAIL-5V@1A-12V@1A-24V@1A-24V@15A
POWERRAIL-5V@1A-12V@1A-24V@1A-24V@16A
The number of cooling fans has not been finalized
GND_RAIL
+24V
E-STOP_NC_CHARGE
T20A
Ideal_Diode_65A
INA226_1
GND_MAIN
+3V3_2
24V
JQX-40F_DC24V@60A_2
DCIN
Control
DCOUT
Control_1
104
50V-2200uF
VIN-
VOUT-
PCBfun_Buck_2
VOUT+
VIN+
Step down
24V -> 14.6
12V Nominal
14.6V Charge
PARRALLE_45Ah
12V_15Ah
50A
Ideal_Diode_100A
12V
should add fuse/diode
before connecting the batts
12V@35 -> 24V@16
25V-1000uF
Voltage_Boost
Change INA226_2's
shunt to external 50A
rated 75mV shunt
JQX-62F_DC12V@80A
Control_3
+5V_2
Tacho
AIR_IN2
AIR_OUT2
COOLING_7
COOLING_8
COOLING_9
+12V
4_Module_Relay
R1_NC
R1_NO
R2_NC
R2_NO
R3_NC
R3_NO
COM
IN1
IN2
IN3
IN4
JQX-40F_DC24V@60A_3
Control_2
12V -> 5V
XL4016_2
24V -> 12V
XL4016_3
XL4016_4
24V -> 5V
XL4016_5
24V -> 19V
+5V_1
+19V
Might change
this into a
more powerful
buck
T5A
T10A
E-STOP_NC_BOT
24V_Main
ESP32-S3_CHARGE
GPIO0
GPIO1
GPIO2
GPIO3
GPIO4
GPIO5
GPIO6
GPIO7
GPIO8
GPIO9
GPIO10
GPIO11
GPIO12
GPIO13
GPIO14
GPIO17
GPIO18
GPIO19/USB_D-
GPIO20/USB_D+
GPIO21
XTAL_32K_P
XTAL_32K_N
XTAL_P
XTAL_N
SPICS1
SPIHD
SPIWP
SPICS0
SPICLK
SPIQ
SPID
SPICLK_N
SPICLK_P
GPIO33
GPIO34
GPIO35
GPIO36
GPIO37
GPIO38
MTCK
MTDO
MTDI
MTMS
U0TXD
U0RXD
GPIO45
GPIO46
CHIP_PU
LNA_IN
This sheet focus only on power,
signal/control wires are not yet
include
Sheet: /OnBoard/
File: Onboard_Power.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 9.0.6
Id: 5/4